

1 / 5 | Ages 7-8 | 20 minutes

Paper bridge test

What we are practising

Change one feature and record an observable result.

What you need

Two equal sheets / Two books of equal height / Five large paper tokens / Pencil

Use small erasers as loads if no young children are nearby.

Before you begin

Put the books on a table with an 8 cm gap; keep this gap fixed.

An adult sets stable books on a low table; use light loads and never stand on a model.

Try it together

1. Lay one sheet flat across the gap and add tokens one at a time at its centre.
2. Record how many it supports before sagging into the gap, or record all five if it holds them.
3. Fold the second sheet into an accordion and repeat with the same gap and tokens.
4. Compare the results and draw the two bridge shapes.

Talk, draw or show your idea

What did you keep the same to make the comparison useful?

Space to explore

2 / 5 | Ages 7-8 | 15 minutes

Where does the water go?

What we are practising

Distinguish absorption from water sitting on a surface.

What you need

Tray / Cloth square / Paper towel square / Plastic lid / Teaspoon / Water

Compare just a cloth and a plastic lid.

Before you begin

Place dry samples on a tray away from electrical devices.

An adult supervises water use; wipe spills promptly and do not drink test water.

Try it together

1. Predict which sample will absorb water.
2. Put one teaspoon of water on each sample.
3. Wait about one minute, then look for spreading, soaking or a puddle.
4. Describe the evidence and dry the tray.

Talk, draw or show your idea

How can you tell soaked-in water from a puddle on top?

Space to explore

3 / 5 | Ages 7-8 | 15 minutes

Make a shadow grow

What we are practising

Relate shadow size to object position while keeping the light fixed.

What you need

Torch / Opaque book / Plain wall

Use a steady desk lamp with an adult positioning it safely.

Before you begin

An adult positions a cool torch facing a wall; keep the room bright enough to walk safely.

Never point a light into anyone's eyes; avoid hot lamps and keep walkways clear.

Try it together

1. Hold the book between the light and wall and observe its shadow.
2. Move the book toward the light without moving the torch.
3. Move the book toward the wall and compare the shadow sizes.
4. Explain which position made the bigger shadow.

Talk, draw or show your idea

What happens if the object blocks more of the spreading light?

Space to explore

4 / 5 | Ages 7-8 | 20 minutes

One place, many living things

What we are practising

Separate direct observations from questions about a habitat.

What you need

Paper / Pencil / Safe view of a garden or potted plant

Use a familiar plant indoors or a picture you already own.

Before you begin

Choose one place observable from a safe path or window.

An adult chooses a safe location; do not touch, taste, collect or feed unknown organisms.

Try it together

1. Spend two quiet minutes looking for plants and animals.
2. Draw three things you see, or draw fewer if only fewer are visible.
3. Label one observed feature, such as a leaf hole or a bird landing.
4. Write a question that your observation alone cannot answer.

Talk, draw or show your idea

Does not seeing an animal prove none live there?

Space to explore

5 / 5 | Ages 7-8 | 15 minutes

Flat or crumpled?

What we are practising

Test the effect of shape on falling through air.

What you need

Two identical sheets of scrap paper

Tear one sheet into two equal halves with adult help.

Before you begin

Clear a floor area and choose a drop height no higher than your shoulder.

Stay on the floor, never climb or drop objects from windows, and retrieve paper before walking.

Try it together

1. Leave one sheet flat and crumple the other tightly.
2. Predict which will land first.
3. Hold them at the same height and release together; repeat three times.
4. Compare results, including any release that was not simultaneous.

Talk, draw or show your idea

Did crumpling add any paper or change only its shape?

Space to explore

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

1. Paper bridge test

A little hint

Use the same number and position of loads for both bridges.

Answer / what to notice

Results vary; a folded sheet often bends less under the same load, but report what actually happened.

Folds change the paper's shape and resistance to bending without adding more paper.

Make it easier

Compare the two bridges with a single token.

Take it further

Repeat each shape three times and compare the results.

Another day, away from the screen

Look for folds or ridges in cardboard packaging.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

2. Where does the water go?

A little hint

Look for a wet patch spreading through the material rather than a movable puddle.

Answer / what to notice

Cloth and paper towel usually soak up water; a solid plastic lid generally keeps it on the surface.

Many absorbent materials have tiny spaces that hold water; a smooth plastic surface does not soak it up in the same way.

Make it easier

Observe only one absorbent and one non-absorbent sample.

Take it further

Compare equal-sized samples of two cloths, repeating with fresh dry pieces.

Another day, away from the screen

Choose which tested material would help wipe up a small spill and explain why.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

3. Make a shadow grow

A little hint

Keep both the torch and wall in the same places.

Answer / what to notice

With a small fixed light source, the book usually makes a larger shadow closer to the light and a smaller one closer to the wall.

Light spreads out from the source, and the opaque object blocks part of that spreading light.

Make it easier

Compare only two marked book positions.

Take it further

Measure shadow width at three positions and record a table.

Another day, away from the screen

Make a short shadow story using your hands.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

4. One place, many living things

A little hint

Quiet or hidden animals may not appear during a short visit.

Answer / what to notice

Any accurate record is useful, including no visible animals; questions should be marked as questions.

A brief observation samples one place and time, so it cannot reveal everything living there.

Make it easier

Describe just one plant and its surroundings.

Take it further

Observe the same spot at a second time and compare without assuming a cause.

Another day, away from the screen

Keep a three-visit observation page and note weather or lighting.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

5. Flat or crumpled?

A little hint

A wide flat sheet pushes against more air as it falls.

Answer / what to notice

The crumpled sheet usually lands first; uneven release or drafts can affect a trial.

The sheets have the same amount of paper, but shape changes air resistance; mass alone does not explain this result.

Make it easier

Watch an adult release both sheets and describe what happened.

Take it further

Compare loose and tight crumpling while keeping paper size the same.

Another day, away from the screen

Sketch a paper shape that might fall slowly, then test it at the same safe height.